A QUANTITATIVE MARKET MICROSTRUCTURE STUDY

Endogenous Liquidity Feedback in a Heterogeneous Mean-Field Market

A numerical study of crowded liquidation and nonlinear price impact

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Research status: theoretical-computational draft; empirical calibration pending.

Individual optimization can become collective liquidity stress when aggregate selling changes the market faced by every trader.

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ABSTRACT

We develop a mean-field game of optimal trading in which a continuum of traders interacts through both the distribution of inventories and aggregate order flow. The principal mechanism is endogenous liquidity: one-sided aggregate flow reduces available liquidity, which magnifies price impact and changes every trader's optimal response. Individual optimization is described by a Hamilton-Jacobi- Bellman equation and the population distribution by a Fokker-Planck equation. The equilibrium is a fixed point in both mean inventory and aggregate trading flow.

We construct a conservative finite-difference solver and validate it against a linear-quadratic benchmark. Grid refinement reduces the maximum mean-inventory error from 0.0259 to 0.0130 while probability is conserved to floating-point precision. In the endogenous-liquidity model, raising the price-feedback coefficient from 0 to 0.35 increases peak selling magnitude by approximately 42%, reduces terminal mean inventory by approximately 18%, and lowers minimum liquidity to 0.468 from a normalized quiet-market level of 1. A multi-start scan over 16 feedback and liquidity-sensitivity combinations finds no evidence of multiple equilibria within the tested region, although the equilibrium response becomes strongly nonlinear. A two-population extension separates fast and slow traders and shows materially different liquidation paths. The results establish a transparent, reproducible platform for studying crowd-induced liquidity stress. They do not yet constitute an empirically calibrated crash model.

INTRODUCTION

Many models of optimal execution describe one strategic trader acting against an exogenous market. That abstraction is useful when the trader is small relative to the market, but it suppresses a central feature of crowded markets: the environment faced by each participant is partly created by the actions of other participants. When many funds hold similar positions, individually reasonable liquidation can produce aggregate selling, weaker liquidity, and more adverse execution conditions.

Mean-field games provide a natural language for this problem. Rather than tracking every pairwise interaction among a large number of traders, a representative trader responds to population-level quantities. The population distribution then evolves under the representative optimal control. Equilibrium requires that the population anticipated by an individual trader agrees with the population generated when all traders optimize.

This paper studies a specific feedback channel. Aggregate trading flow determines a smooth liquidity index. The same flow produces permanent price drift inversely proportional to liquidity. A trader who remains long during aggregate selling bears an adverse mark-to-market effect and therefore has an additional reason to sell. The resulting loop can amplify order flow even though no trader is programmed to panic.

The contributions of this draft are fourfold:

- We formulate a coupled mean-field game in which both mean inventory and aggregate control must satisfy equilibrium consistency conditions.
- We implement a conservative HJB-Fokker-Planck solver and validate it against a solvable linear-quadratic benchmark.
- We quantify nonlinear amplification and perform a multi-start parameter scan that distinguishes amplification from numerical evidence of equilibrium multiplicity.
- We extend the framework to heterogeneous fast and slow populations and provide a reproducible calibration interface for future market data.

The present contribution is primarily methodological and computational. The chosen liquidity function is a transparent hypothesis rather than a claimed universal law. Empirical calibration and formal existence or uniqueness proofs remain future work.

RELATED LITERATURE

The work is closest to mean-field models of trade crowding and optimal execution. Cardaliaguet and Lehalle [1] formulate optimal liquidation as a mean-field game of controls in which aggregate trading affects public prices and heterogeneous risk preferences can be represented. Casgrain and Jaimungal [2] study heterogeneous algorithmic traders with partial information, permanent and temporary price impact, and a vector-valued forward-backward characterization. Neuman and Voss [3] analyze crowded liquidation under transient price impact and prove convergence from the finite-player game to the mean-field limit.

Our focus on endogenous liquidity also relates to equilibrium descriptions of the limit order book. Ma and Noh [4] model liquidity using a nonlocal reflected mean-field stochastic differential equation and connect the order-book frontier to a value function. Lachapelle, Lasry, Lehalle, and Lions [5] use a mean-field-game analysis of heterogeneous institutional and high-frequency participants to study price formation, liquidity imbalance, and spreads. Fujii and Takahashi [6] derive equilibrium price formation in a market with a major firm and a mean field of minor firms.

The distinguishing feature of the present model is not the use of mean-field games or price impact alone. It is the explicit feedback from aggregate control to a bounded liquidity state and then to the price drift entering each trader's running objective, combined with a numerical search for nonlinear response, initial-guess dependence, and heterogeneous population effects.

HOMOGENEOUS-AGENT MODEL

State dynamics

The state of a representative trader is inventory Q_t. The trader controls the trading rate u_t:

$$dQ_t = u_t\,dt + \sigma\,dW_t.$$

Negative u denotes selling. The Brownian component represents idiosyncratic inventory disturbances such as unmodelled fills or client flow. It also regularizes the population density. In a pure execution interpretation, this noise may instead be placed in fills or signals; the present placement is a modelling simplification.

Let m(t,q) denote the population density over inventories. Its mean is

$$\mu_t = \int q\, m(t,q)\, dq.$$

Aggregate order flow

If a trader at state q uses feedback control u*(t,q), aggregate order flow is

$$A_t = \int u^*(t,q) m(t,q)\, dq.$$

Negative A denotes net selling. This is a mean field of controls: interaction occurs through the population's actions as well as its states.

Endogenous liquidity and price drift

Liquidity is postulated to be

$$L(A) = L_{min} + (L_0 - L_{min})/(1 + \chi A^2),$$

where L_0 is quiet-market liquidity, L_{min} is a positive floor, and χ controls sensitivity to one-sided flow. The price-impact drift is

$$b(A) = \beta A/L(A),\ dS_t = b(A_t)dt.$$

The bounded liquidity floor prevents a singular division by zero. This form is smooth, symmetric between buying and selling, and deliberately parsimonious. It is not intended to settle the empirical form of liquidity depletion.

Trader objective

The running cost is

$$\ell(q,u,m,A) = \tfrac{1}{2}\eta u^2 + \tfrac{1}{2}\gamma q^2 + \tfrac{1}{2}\kappa(q - \mu_t)^2 - qb(A_t).$$

The first term captures temporary trading cost, the second inventory risk, and the third pressure to remain near the population mean. The final term is the negative of expected mark-to-market gain. If A is negative, b(A) is negative. For a long trader, -qb(A) is positive, so remaining long during population selling is costly.

The objective is

$$J(u) = E[\int_0^T \ell(Q_t,u_t,m_t,A_t)dt + \tfrac{1}{2}GQ(T)^2].$$

HJB-Fokker-Planck system

Dynamic programming gives the minimized HJB equation

$$-\partial_t V = \tfrac{1}{2}\gamma q^2 + \tfrac{1}{2}\kappa(q - \mu_t)^2 - qb(A_t) - (\partial qV)^2/(2\eta) + \tfrac{1}{2}\sigma^2\partial qqV,$$

with terminal condition $V(T,q)=Gq^2/2$. The optimal control is

$$u^*(t,q) = -\partial qV(t,q)/\eta.$$

The population density follows

$$\partial_t m = -\partial q(um) + \tfrac{1}{2}\sigma^2\partial qqm.$$

Equilibrium requires the two consistency conditions

$$\mu_t = \int q\, m(t,q)dq,\ A_t = \int u^*(t,q)m(t,q)dq.$$

The feedback loop is therefore A → L → b → V → u* → A.

LINEAR-QUADRATIC BENCHMARK

Before solving the full grid problem, we remove price feedback and exploit the quadratic structure. For the ansatz

$$V(t,q) = \tfrac{1}{2}a_t q^2 + b_t q + c_t,$$

the optimal control is linear in inventory. A Gaussian initial population remains Gaussian, so the system reduces to backward equations for a and b and forward equations for the population mean μ and variance s:

$$a'_t = a_t^2/\eta - (\gamma + \kappa),\ b'_t = \kappa\mu_t + a_t b_t/\eta,$$

$$\mu'_t = -(a_t\mu_t + b_t)/\eta,\ s'_t = -2a_t s_t/\eta + \sigma^2.$$

This benchmark supplies a transparent fixed point and a numerical reference for the general PDE solver. Under the default parameters, mean inventory falls from 1.000 to approximately 0.311.

NUMERICAL METHOD

Discretization

Inventory is discretized on a uniform grid qj with spacing Δq, and time on tn with spacing Δt. Centered differences approximate the first and second derivatives in the HJB. Starting from the terminal condition, the HJB is stepped backward. The optimal feedback is recovered from the discrete inventory gradient.

The Fokker-Planck equation is written as a conservation law with flux

$$J = um - \tfrac{1}{2}\sigma^2 \partial_q m.$$

Advection uses an upwind face value and diffusion uses a centered difference. Zero total probability flux is imposed at the truncated inventory boundaries. The method is conservative by construction. An explicit CFL check rejects parameter-grid combinations that could move too much probability in one step.

Fixed-point iteration

For the homogeneous feedback model, the algorithm is:

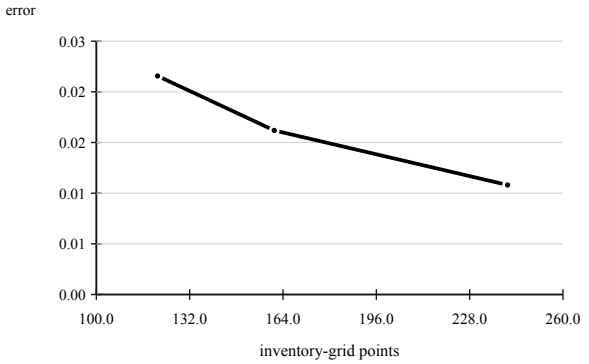
1. Guess complete paths μ(t) and A(t).
2. Calculate liquidity and price drift.
3. Solve the HJB backward.
4. Calculate u* on the grid.
5. Solve the Fokker-Planck equation forward.
6. Recalculate μ(t) and A(t).
7. Relax the old and generated paths and repeat.

Convergence is declared when the maximum residual across both paths is below the chosen tolerance.

Verification

Probability mass is conserved to approximately 10^{-15} and numerical density remains nonnegative in the reported experiments. Refinement from 121 to 241 inventory points reduces maximum mean-path error against the moment benchmark from 0.0259 to 0.0130. The error pattern is consistent with the first-order upwind treatment of transport.

EXHIBIT 1 - GRID REFINEMENT



Mean-path error decreases as the numerical grid is refined.

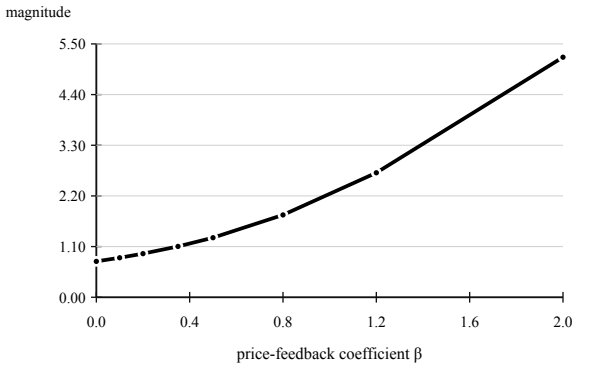
HOMOGENEOUS FEEDBACK RESULTS

One-dimensional feedback sweep

We vary β with all other illustrative parameters fixed. At β=0, terminal mean inventory is 0.331 and peak selling flow is -0.779. At β=0.35, terminal mean falls to 0.272 and peak selling becomes -1.103. Thus selling magnitude increases by roughly 42%, while terminal inventory falls by roughly 18%. Minimum liquidity falls to 0.468 and cumulative price impact is -0.458 in normalized units.

At β=2.0, the model produces peak selling of -5.212, minimum liquidity of 0.264, and cumulative impact of -5.002. These magnitudes should not be interpreted as empirical forecasts; their role is to reveal the nonlinear internal response of the stylized system.

EXHIBIT 2 - NONLINEAR AMPLIFICATION



Peak selling magnitude rises nonlinearly with price feedback.

Multi-start bifurcation scan

We evaluate 16 combinations of β in {0, 0.35, 0.8, 1.2} and χ in {0, 1, 2, 4}. Each pair is solved from mild, baseline, and severe initial flow paths. The largest spread in terminal mean inventory across initializations is 2.63×10^{-7}, below the fixed-point tolerance used for the scan.

Accordingly, this experiment provides no evidence of multiple equilibria in the sampled parameter region. It does show increasingly nonlinear amplification. This distinction matters: a steep response curve is not by itself a mathematical phase transition. Wider scans, continuation methods, fixed-point Jacobian estimates, and formal analysis are required before making a multiplicity claim.

HETEROGENEOUS FAST AND SLOW TRADERS

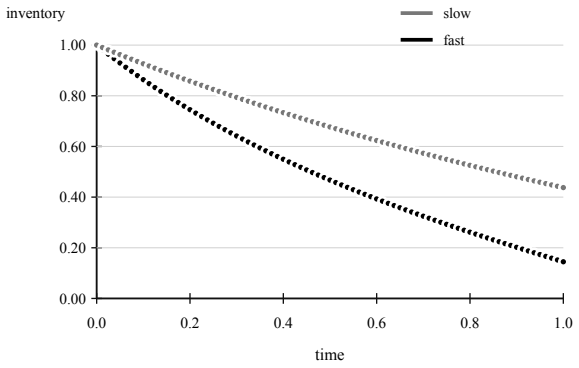
The two-population extension assigns separate value functions, controls, densities, and parameters to fast and slow traders. Both populations respond to the same price drift and aggregate mean. Total flow is

$$A_t = wF\int uF\, mF\, dq + wS\int uS\, mS\, dq,\ wF + wS = 1.$$

The illustrative configuration gives fast traders lower trading cost, higher inventory aversion, and a stronger terminal penalty. Their population weight is 0.35. Slow traders have weight 0.65 and trade more cautiously.

The equilibrium converges in 99 iterations. Terminal mean inventory is 0.145 for fast traders and 0.438 for slow traders; the weighted population mean is 0.335. Peak aggregate selling is -1.037 and cumulative price impact is -0.398. The result shows meaningful separation in liquidation paths, but it does not yet establish whether heterogeneity stabilizes the market relative to a properly matched homogeneous counterfactual. That comparison is an explicit next experiment.

EXHIBIT 3 - HETEROGENEOUS TRADERS



Fast traders liquidate more aggressively than slow traders.

CALIBRATION DESIGN

The repository includes a data-source-neutral calibration interface. Given observed aggregate flow A and short-horizon price drift y, it estimates β and χ by minimizing

$$MSE(\beta,\chi) = N^{-1}\,\Sigma_i\,[\,y_i - \beta A_i / L(A_i;\chi)\,]^2.$$

A deterministic synthetic test recovers β=0.40 and χ=1.50 exactly on the supplied parameter grid. This verifies the calibration code but is not empirical validation.

A future data adapter should construct synchronized observations of signed order-flow imbalance, short-horizon mid-price changes, spread, and displayed depth. Public cryptocurrency data is a practical initial candidate because continuous trading and order-book updates are accessible. Estimation must separate model fitting and evaluation windows and compare against simpler linear-impact and fixed-liquidity baselines.

REPRODUCIBILITY

All reported results are generated by repository scripts. The codebase separates the analytical benchmark, general grid solver, homogeneous liquidity model, heterogeneous model, calibration routine, and experiments. Automated tests cover mass conservation, density positivity, terminal conditions, recovery of the analytical benchmark, recovery of the no-feedback model, direction of the feedback effect, weighted-population consistency, and synthetic parameter recovery.

The present suite contains 20 passing tests. CSV outputs preserve the paths and summary statistics used for figures. Randomness is absent from the deterministic PDE experiments; future Monte Carlo or empirical routines will record explicit seeds.

LIMITATIONS

Several limitations prevent the current draft from being interpreted as a completed empirical asset-pricing result.

- The endogenous liquidity curve is imposed rather than derived from an explicit limit-order-book equilibrium.
- Inventory noise is a reduced-form representation of idiosyncratic fills or flows.
- The price equation currently contains deterministic endogenous drift but no common market noise.
- Parameter values are normalized and illustrative.
- The numerical scan does not prove existence, uniqueness, or nonexistence of other equilibria outside the tested region.
- The first-order upwind scheme introduces numerical diffusion.

- The heterogeneous result has not yet been compared with a moment-matched homogeneous counterfactual.
- No real-market calibration or out-of-sample test is reported.

These limitations define the next research program rather than being hidden behind the numerical results.

CONCLUSION

We introduced an endogenous-liquidity mean-field game in which aggregate optimal trading changes liquidity and price drift, feeding back into individual decisions. The model is solved with a conservative backward-forward numerical method validated against a linear-quadratic benchmark. The experiments establish robust nonlinear selling amplification under the selected mechanism. A multi-start scan finds no evidence of multiple equilibria in the initial parameter region, while a two-population extension produces distinct fast and slow liquidation paths.

The draft's main value is a transparent and reproducible foundation. The next steps are a wider continuation-based bifurcation analysis, matched heterogeneity counterfactuals, alternative liquidity specifications, and empirical calibration. Only after those steps should the model be described as evidence about real market crashes rather than a mathematically controlled mechanism experiment.

REFERENCES

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APPENDIX A. DEFAULT HOMOGENEOUS PARAMETERS

The normalized baseline uses horizon $T=1$, temporary cost $\eta=1$, inventory penalty $\gamma=0.25$, crowding $\kappa=0.75$, terminal penalty $G=2$, inventory noise $\sigma=0.20$, initial mean 1, and initial variance 0.16. The liquidity extension uses $L_0=1$, $L_{min}=0.25$, $\chi=2$, and default $\beta=0.35$.